Targeted Active Cooling for UK Homes

A plain-English proposal for public health teams, construction companies, housing providers, employers and business owners

Recommendation: targeted cooling where risk remains

- Prioritise vulnerable people and high-risk homes
- Use shading and ventilation first
- Install efficient active cooling where passive measures are not enough
- Monitor comfort, bills, carbon and maintenance

Home Heat Risk AI | May 2026 | Draft report for discussion and stakeholder engagement

1. Executive summary

Plain-English summary: The UK is getting hotter, and many homes were designed to keep warmth in, not let heat out. This report proposes a targeted programme to install high-efficiency active cooling - including air conditioning or reversible heat-pump cooling - in the homes and buildings where passive measures are no longer enough.

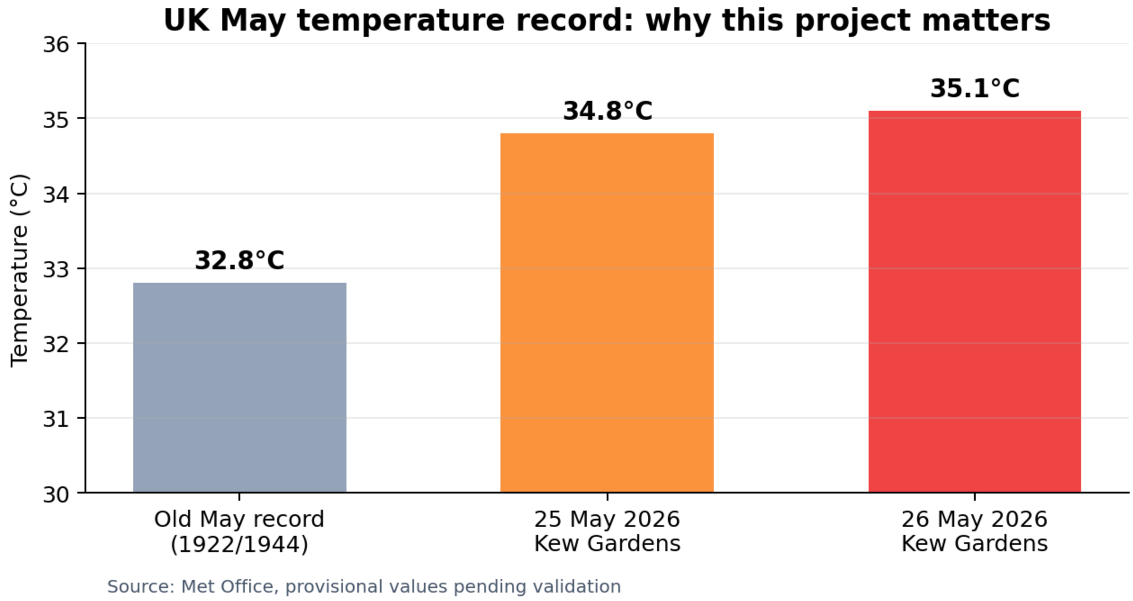
What is the problem?	Heat is becoming a public health and business continuity issue, not just a comfort issue.
Who is most at risk?	Older people, people with health conditions, young children, pregnant people, low-income renters
What is proposed?	A risk-based cooling programme: shade and ventilation first, then high-efficiency active cooling fo
What should not happen?	A blanket, unmanaged rush to cheap portable AC that raises bills, strains the grid and ignores ve
What should happen?	Public health triage, building surveys, high-efficiency systems, electricity planning, tenant protecti

Recommended policy position

This report does **not** recommend uncontrolled mass installation of air conditioning in every home. It recommends a **targeted active-cooling pathway**: use passive cooling where it works; install high-efficiency active cooling where health, building form and heat exposure show that passive measures are unlikely to be enough.

2. Why this is urgent now

In late May 2026, the UK provisionally broke its May and spring temperature record on two consecutive days: 34.8°C at Kew Gardens on 25 May and 35.1°C at Kew Gardens on 26 May. The Met Office also reported unusually warm nights, including a May tropical night where temperatures stayed above 20°C. [1][2]



Record heat matters because homes do not cool down instantly. Warm nights reduce the body's recovery time and make overheated homes more dangerous.

The Climate Change Committee projects that, without additional adaptation, 92% of existing UK homes could be at risk of overheating by 2050 under a 2°C global warming level. It also states that most future heat-related mortality risk could be reduced by adapting the most vulnerable urban households, and that active cooling is likely to have a role alongside passive measures. [3]

Simple explanation: A warm home is like a bottle of hot water. If the night is cool, it can cool down. If the night stays warm, the heat remains trapped inside.

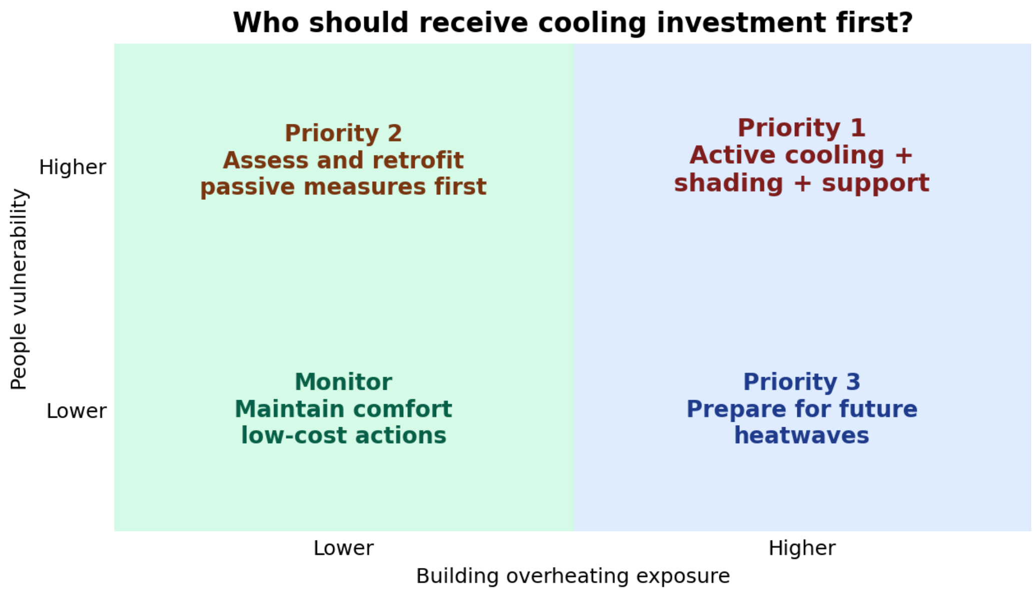
3. Heat is a public health issue

Heat-health alerts are already used in England because heat can increase deaths, increase pressure on health and social care services, affect medicines, disrupt staffing and make indoor environments unsafe for vulnerable people. UKHSA heat-health alert pages explicitly warn that indoor environments may overheat and that vulnerable people are at greater risk. [4][5]

Heat outside	Homes absorb heat	Nights stay warm	Bodies cannot recover	Health and work impacts
High daytime temperatures	Sun through windows, roof heat, trapped air	No overnight cooling	Sleep loss, dehydration, cardiac stress	More demand on care, lost productivity

Who should be prioritised?

Priority should be based on both **building risk** and **people risk**. A low-income older tenant in a top-floor flat may need help before an affluent household that can buy a system independently.



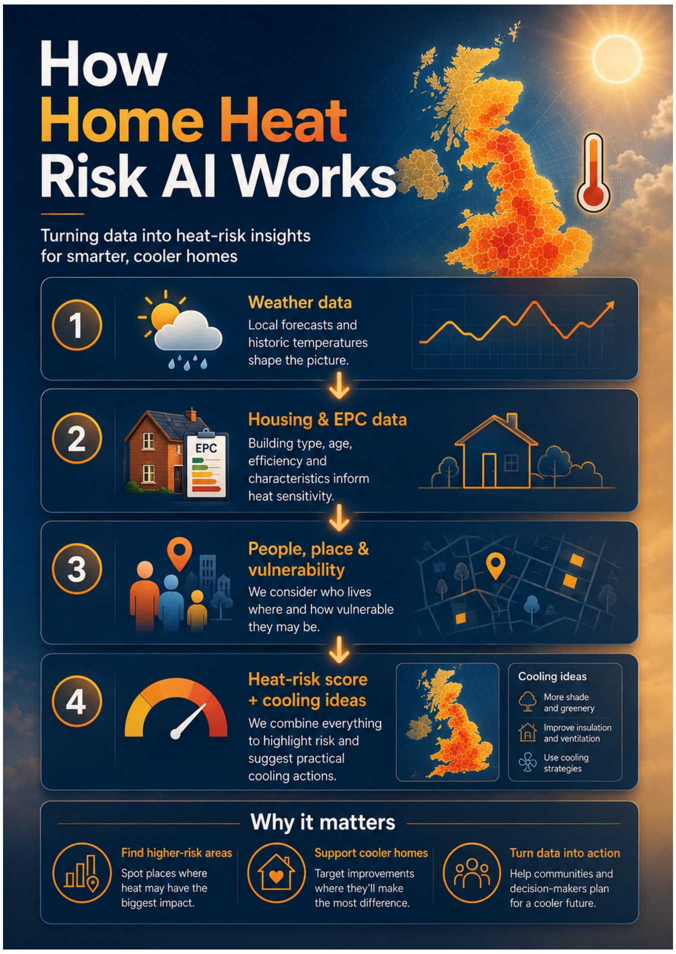
4. Why UK homes can overheat

Many UK homes are good at keeping heat in during winter. That is valuable for cold weather and energy efficiency. But during a heatwave, the same features can trap unwanted heat, especially in flats with limited cross-ventilation or high solar gain.

Risk factor	Plain-English explanation	Practical response
Top-floor flats	Heat rises and roofs absorb sunlight.	Roof/loft measures, shading, ventilation, active cooling if neede
Single-aspect homes	Windows on one side only make it harder to create a breeze.	Secure night ventilation, mechanical ventilation, cooling suppor
Large unshaded windows	Sunlight acts like a heater through glass.	External shading, blinds, films, trees, balconies, overhangs.
Dense urban areas	Concrete, roads and buildings store heat.	Urban greening, cool roofs, shaded streets, cooling centres.
Low-income renting	Residents may lack permission or money to adapt.	Landlord standards, grants, tenant protections, targeted suppor

Approved Document O already sets overheating mitigation standards for new residential buildings in England, but it is mainly a new-build regulation. Existing homes still need retrofit strategies, especially where vulnerable people live. [6]

5. How Home Heat Risk AI helps

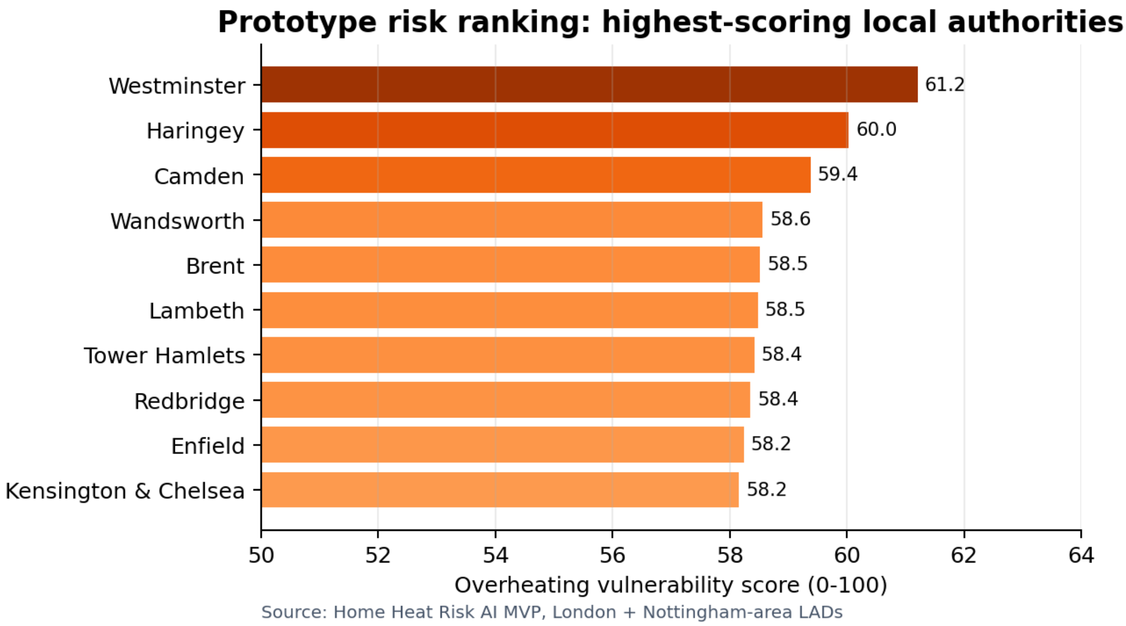


The tool is a decision-support prototype: it turns several public datasets into a simple overheating vulnerability score and cooling recommendation.

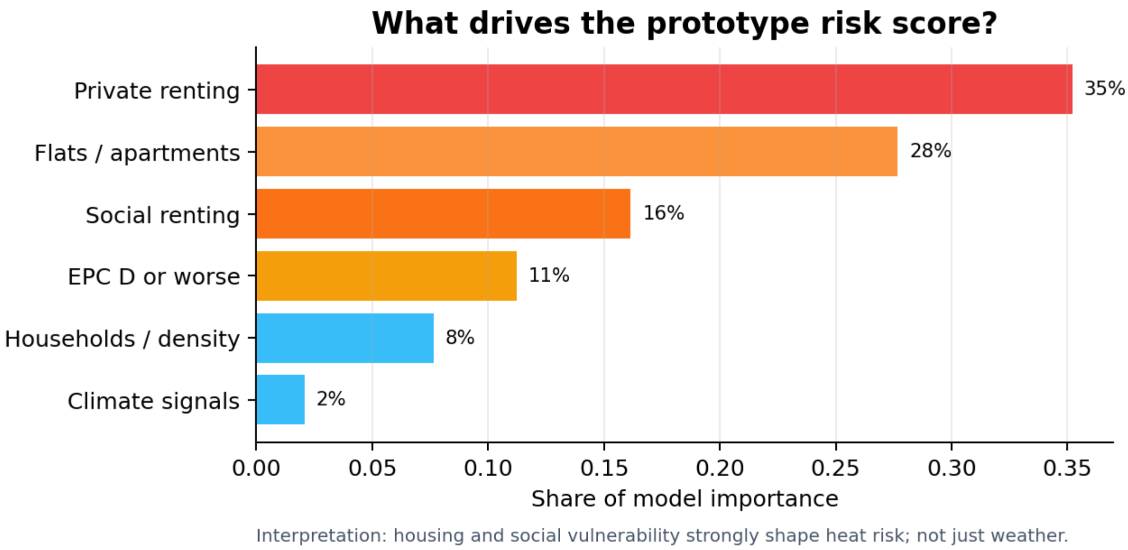
Important: The score is not a medical diagnosis and it does not measure exact indoor temperature. It is a screening tool that helps decide where to look first.

6. What the prototype found

The MVP covers 41 London and Nottingham-area local authorities. It ranks areas by an explainable overheating vulnerability score. The highest-scoring areas in the current prototype are dense London boroughs, which is plausible because many risk drivers - flats, renting, density and limited cooling options - are concentrated there.

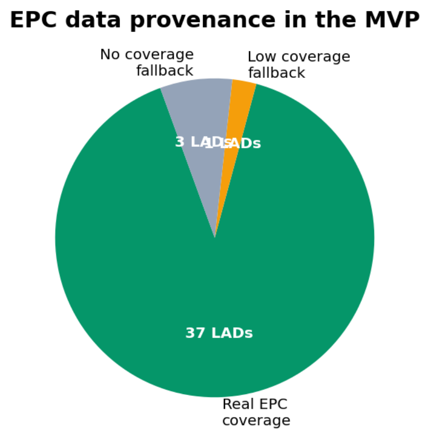


The model interpretation also shows that the story is not only about the weather. Housing and social factors matter strongly. This is why an air-conditioning policy must be targeted and fair, not simply left to household purchasing power.



7. Data quality and honesty

A trustworthy public-facing tool should show where its data is strong and where it is approximate. The Home Heat Risk AI prototype now uses a hybrid EPC layer: real EPC records where there is enough local coverage, and a labelled fallback where coverage is missing or too low.



What the EPC labels mean

Real EPC: enough EPC certificates are available for that local authority.

Low-coverage fallback: fewer than 500 valid certificates; use a statistical fallback instead of trusting a tiny sample.

No-coverage fallback: the source data does not cover that area, so the EPC layer is approximate.

Why this matters: A decision tool that labels weak data is more useful than one that hides uncertainty. Public bodies, developers and landlords need to know when a recommendation is based on strong evidence and when a site survey is essential.

8. Proposal: targeted installation of active cooling

The central proposal is a **risk-based active-cooling programme**. This means installing high-efficiency active cooling where the combination of heat exposure, building form and vulnerability shows that passive measures are unlikely to protect people adequately.

Cooling hierarchy: do simple things first, but do not stop there when risk remains high

1	**Reduce heat entering**	External shading, tree shade, reflective surfaces, window films where suitable.
2	**Move heat out**	Safe night ventilation, cross-ventilation, fans, mechanical ventilation where appropriate.
3	**Protect vulnerable people**	Cool rooms, check-ins, grants, public health triage, cooling centres.
4	**Install efficient active cooling**	Reversible heat-pump cooling or high-efficiency fixed AC in priority homes and buildings.
5	**Operate responsibly**	Smart controls, maintenance, low-carbon electricity, demand management, user guidance

Preferred technology direction

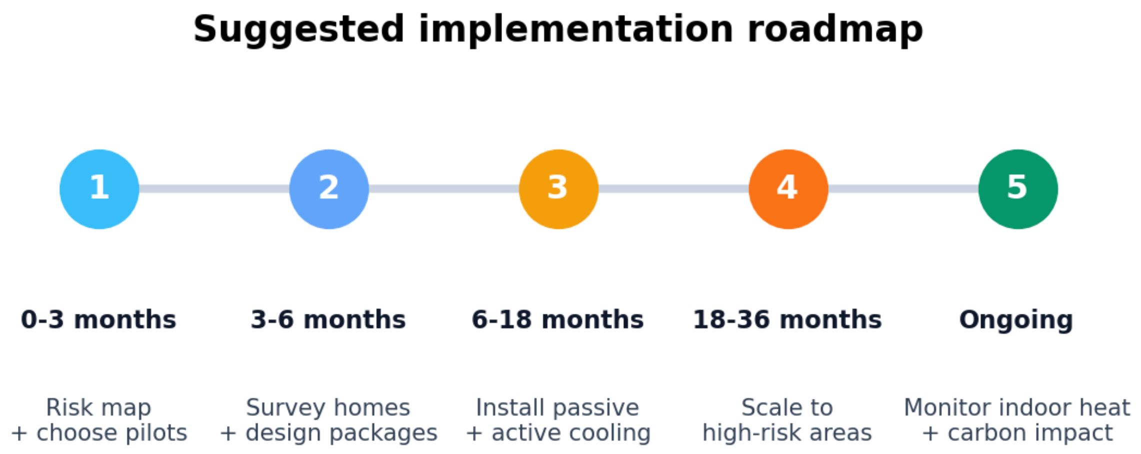
Where active cooling is needed, the preferred option should usually be **high-efficiency fixed systems**, such as reversible air-to-air heat-pump cooling, rather than unmanaged portable units. Fixed systems are more controllable, can be specified professionally, can support winter heating strategies in some cases, and can be managed through maintenance and energy standards.

9. What different audiences should do

Audience	Main decision	Immediate action
Public health bodies (UKHSA/OHID, formerly PHE)	Treat overheated homes as a health protection issue.	Create heat-vulnerability lists, target outreach, support indoor heat monitoring, advise on priority groups.
Local authorities	Use data to prioritise neighbourhoods and vulnerable housing.	Map high-risk flats, social housing, private rented homes and care settings.
Construction companies	Design for future summers, not past summers.	Use dynamic overheating assessment, shading, ventilation and active-cooling-ready design.
Housing providers and landlords	Protect tenants and avoid cooling poverty.	Survey top-floor/single-aspect flats, install shading and cooling packages in priority units.
Business owners	Heat affects customers, staff and productivity.	Assess workplace heat risk, cooling demand, staff welfare, customer comfort and business continuity.
Energy networks and suppliers	Plan for cooling demand before it arrives chaotically.	Support smart controls, demand shifting and low-carbon cooling tariffs.

Fool-proof rule: If a building regularly becomes too hot for vulnerable people to sleep, work, recover or receive care safely, then passive measures alone may not be enough. That building needs an active-cooling assessment.

10. Implementation roadmap



A practical programme should start small, prove the method, and then scale. The first pilots should focus on areas with high heat risk and high vulnerability: top-floor flats, care settings, social housing, private rented homes with low adaptive capacity, and small businesses or workplaces where heat affects staff welfare.

Stage	Checklist
Risk mapping	Use heat, housing, EPC, tenure and vulnerability data to rank areas.
Site surveys	Confirm building layout, solar gain, ventilation, electrical capacity, noise, fire safety and resident needs.
Package selection	Choose passive-only, passive-plus, or active cooling based on the risk matrix.
Procurement	Specify high-efficiency systems, safe installation, maintenance access and user guidance.
Monitoring	Track indoor temperatures, user comfort, bills, demand peaks and health/service outcomes.

11. Why this matters for businesses and construction

Active cooling is not only a household comfort issue. It is also a productivity, safety, reputational and operational resilience issue. Businesses that ignore overheating risk may face staff wellbeing issues, reduced customer comfort, equipment risk, lower productivity and higher emergency costs during heatwaves.

Sector	Heat risk	Opportunity
Construction and development	New buildings may be criticised if they overheat soon after completion.	Design homes that are cooling-ready, resilient and compliant with evolving expectations.
Care and health settings	Heat can disrupt care, medication management, staffing and patient comfort.	Prioritise clinical-risk spaces and vulnerable residents.
Retail and hospitality	Hot premises reduce comfort, dwell time and staff wellbeing.	Use efficient cooling to maintain service quality and customer experience.
Offices and education	Heat affects concentration, attendance and performance.	Target spaces used during heat-health alerts and exam/work periods.
Housing providers	Tenants may face unsafe indoor heat and cooling poverty.	Use fair, evidence-based retrofit and active-cooling programmes.

Commercial message

Cooling should be treated as a planned infrastructure investment, not an emergency purchase made during a heatwave. Planned systems can be efficient, maintained, properly sized and integrated with net-zero strategies.

12. Guardrails: how to avoid bad cooling

Installing cooling without guardrails can create new problems: high bills, grid peaks, noise, poor maintenance, overheating of neighbours, refrigerant impacts, and unfair access. A good programme must include these protections.

Guardrail	What it means in practice
Equity first	Prioritise vulnerable people and those who cannot afford their own adaptations.
Passive-first	Shade, ventilation and fabric measures should be assessed before or alongside active cooling.
Efficient equipment	Avoid cheap, inefficient emergency purchases where fixed high-efficiency systems are possible.
Low-carbon operation	Use smart controls, efficient settings and demand management to reduce grid stress.
Data privacy	Indoor monitoring should be consent-based, minimal and used for safety, not surveillance.
Maintenance	Cooling systems must be serviceable; a broken AC unit is not a resilience plan.

Public trust point: The message should not be 'everyone must buy AC'. The message should be 'no one should be left in unsafe indoor heat because their home was not designed for the climate we now experience'.

13. Simple decision checklist

Use this checklist before approving, funding or installing active cooling.

Question	Yes / No
Is the home/building in a high heat-risk area?	
Is it a top-floor, single-aspect, highly glazed or poorly ventilated space?	
Do vulnerable people live, work, study or receive care there?	
Have passive measures been assessed and found insufficient or incomplete?	
Is the proposed cooling system efficient, maintainable and professionally installed?	
Is there a plan for bills, user guidance, maintenance and monitoring?	
Can the system operate without creating unreasonable noise or neighbour impacts?	

Suggested success indicators

1. number of high-risk homes assessed; **2.** number of vulnerable households protected; **3.** share of systems meeting efficiency standards; **4.** indoor overheating hours reduced; **5.** tenant/user satisfaction; **6.** electricity demand managed during heat alerts; **7.** maintenance completion rate.

14. Limitations and responsible use

This report is a proposal and decision-support document. It should not replace professional building surveys, public health judgement, HVAC engineering advice, fire safety review, planning requirements or resident consultation.

Limitation	What to do about it
Area-level model	Use it to prioritise inspections; do not assume every home in an area has the same risk.
No direct indoor temperature data	Add indoor sensors in pilot homes with informed consent and privacy safeguards.
EPC coverage varies	Keep EPC source labels visible and require site surveys for low-coverage areas.
Climate data resolution	Upgrade to daily 5km HadUK-Grid hot-day and tropical-night metrics when available.
Behaviour matters	Resident guidance and support are part of the intervention, not an afterthought.

Note on Public Health England

Public Health England formally closed in 2021, with health protection functions transferred to the UK Health Security Agency and health improvement functions to bodies including the Office for Health Improvement and Disparities. This report uses 'public health bodies' to refer to the current landscape, while recognising that many stakeholders still use the PHE name informally. [7]

15. References

[1]	Met Office (25 May 2026). Provisional spring daily temperature record as heatwave continues. 34.8°C at Kew Gardens.
[2]	Met Office (26-27 May 2026). UK May and spring temperature record provisionally broken for second day in a row; Deep Dive: How unusual is the heat? 35.1°C at Kew Gardens and first recorded May tropical night.
[3]	Climate Change Committee (2026). A Well-Adapted UK - Fourth Independent Assessment of UK Climate Risk. Homes overheating risk, vulnerable households, active and passive cooling.
[4]	UKHSA Weather-Health Alerts dashboard (May 2026). Heat-health alerts for London and other English regions.
[5]	UKHSA Weather-Health Alerts dashboard. Heat-health alert risk descriptions: increased deaths, healthcare demand, indoor overheating, care settings and workforce impacts.
[6]	GOV.UK. Overheating: Approved Document O. Building regulation guidance for overheating mitigation in new residential buildings in England.
[7]	GOV.UK (1 October 2021). Public health system reforms: location of Public Health England functions from 1 October 2021.

Final message: Cooling is no longer a luxury discussion. It is becoming a public health, housing quality, design, business continuity and fairness discussion. The safest path is a planned, targeted and evidence-led active-cooling programme.